

 quarter = 4 equal size pieces.

1 litre = 1.06 ^{cuarto} quarts

4 quarts = 1 gallon

rounded to 10th how many litres per 1 gallon.

$0.943 \text{ litre} = 1 \text{ quart}$

$3.77 \text{ litre} = \text{gallon}$

4 quarts = 1 gallon

"a" litres = 1 quart
1 litre = 1.06 quarts

$$a = \frac{1}{1.06} = 0.943$$

how much = 4.24 quarts.
1.06 quarts = 4L
1.06 quarts = 3L
1.06 quarts = 2L
1.06 quarts = 1L

1 gallon

1 litre = 1.06 quarts
a litre = 1 quart

$$\begin{aligned} \text{a litre} &= \frac{1 \text{ quart}}{1.06} \\ \text{a litres} &= \frac{1}{1.06} \\ &= 0.943 \text{ litres} \end{aligned}$$

in 1 quart
4 quarts = 1 gallon

$$\begin{aligned} 4 \text{ litres} &= 4 \times 0.943 \\ &= 3.77 \text{ litres (gallon)} \end{aligned}$$

To make a drink

6-8 fl oz
1 pouch

6-8 fl oz per pouch mix

3 cups of water to 4 pouches of mix

1 cup = 8 fl oz
1 cup = 8 fl oz
1 cup = 8 fl oz
24 fl oz
4 pouches of mix

how much water do I need for 4 pouches?

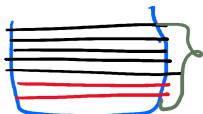
1 cup = 8 fl oz

fl = Fluid = liquid
oz = measure = volume.

Unit	Sample Use	Comparing Units
fluid ounce (fl oz)	 1 fl oz	1 fl oz = $\frac{1}{8}$ c
cup (c)	 1 cup 8 fl oz	1 c = 8 fl oz
pint (pt)	 Milk	1 pt = 2 c
quart (qt)	 motor oil	1 qt = 2 pt
gallon (gal)		1 gal = 4 qt

$$\begin{aligned} 24 \text{ fl oz of water} &= \frac{6 \text{ fl oz}}{\text{pouch of mix}} \\ 4 \text{ pouches of mix} & \end{aligned}$$

1 gallon = 3.77

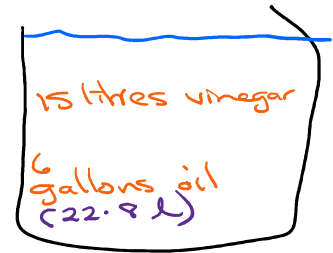


3 x amount

oil : vinegar
2 gallons + 5 litres $\times 3$ 22.641 litres
6 gallons + 15 litres
22.8

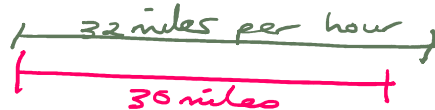
3x amount $\swarrow$ 2 gallons = 5 litres $\swarrow \times 3$ 22.641 litres 6 gallons (oil)
 6 gallons = 15 litres $\swarrow \times 3$ 22.8
 $22.8 \text{ l} = 15 \text{ l}$

Total litres = $15 + 22.8$
 $= 37.8 \text{ litres.}$



Oil Vinegar
 2 gallons = 5 litres
 3x amount $\swarrow \times 3$ 7.54 litre = 5 litres $\swarrow \times 3$
 $22.62 = 15 \text{ litres}$

drive 32 mph average place is 30 miles away
 How long to get there?



miles hour
 $32 : 1$
 $30 : x$

hours miles
 $\frac{x}{1} : \frac{30}{32}$

$x = \frac{30}{32}$
 $= \frac{15}{16} = 0.9375$
 Round to 1 hour?

$y \text{ mins} = 0.9375 \text{ hours}$
 $60 \text{ mins} = 1 \text{ hour}$

$60 \times \frac{y}{60} = 0.9375 \times 60$
 $y = 0.9375 \times 60$
 $= 56.25 \text{ mins}$

1. \$150 10% off

$150 = 100\%$
 $15 = 10\%$

$150 - 15 = \$135$



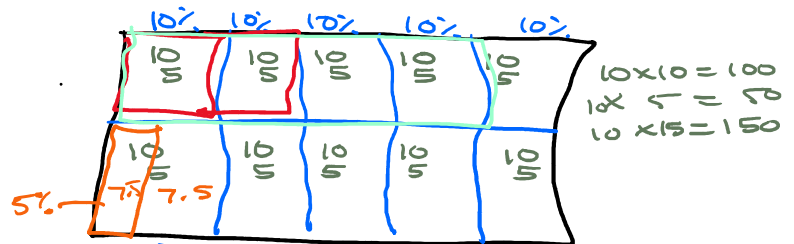
$150 \times 0.9 = 135$

$(150 \times 0.9 = 135)$

$150 \times 1 = 150$

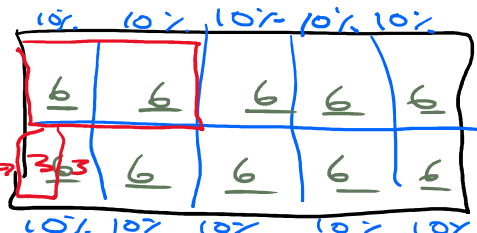
2. What is 10% of \$135

10% of 135 = \$13.50



20% of 150 = 30 5% of 150 = 7.5
 40% of 150 = 60

\$60



$90\% = \frac{90}{100}$
 $= \frac{9}{10}$
 $= 0.9$

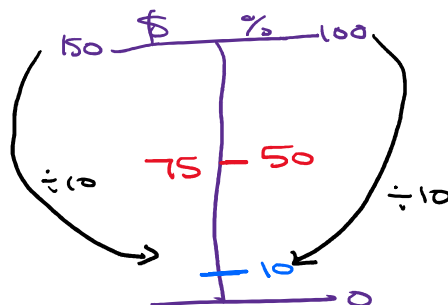
$100\% - 10\% = 90\%$ 10% of 60 = 6
 $60 - 6 = 54$ 20% of 60 = 12
 5% of 60 = 3 $\rightarrow$ 3+3 = 6
 (10% off) = 90% of 60 = 54

$60 - 6 = 54$
 $20\% \text{ of } 60 = 12$
 $5\% \text{ of } 60 = 3$
 $10\% \text{ off} = 90\%$
 $60 = 54$

33	6	6	6	6
10%	10%	10%	10%	10%

$0 = 00\%$
 $\frac{1}{10} = 0.1$
 0.9
 0.90%

	10%	10%	10%	10%	10%
10%	15	15	15	15	15
10%	15	15	15	15	15



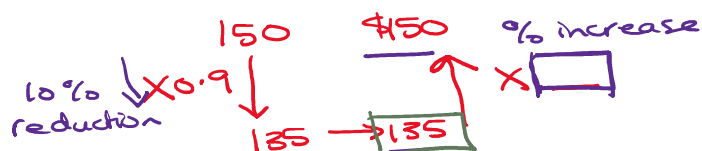
$$\begin{array}{r} 2 \begin{array}{|c|c|} \hline x & -3 \\ \hline 2x & -6 \\ \hline \end{array} \\ 3 \begin{array}{|c|c|} \hline x & +2 \\ \hline 3x & +6 \\ \hline \end{array} \end{array}$$

$$\begin{aligned} \underline{2(x-3)} + 4 &= \underline{3(x+2)} - 5 \\ \underline{2x-6} + 4 &= \underline{3x+6} - 5 \\ 2x - \overset{-2x}{2} &= \overset{-2x}{3x} + 1 \\ -2 &= x + 1 \\ -3 &= x \\ x &= -3 \end{aligned}$$

$$\begin{aligned} 2x - 2 &= 3x + 1 \\ 2x - 2x &= 3x - 2x + 3 \\ 0 &= x + 3 \\ -3 &= x \end{aligned}$$

Expand (get rid of the brackets)

Shop sells \$150, 10% off everything
new price = \$135 $(150 \times 0.9 \text{ or } 150 - 15)$
 (90%) $100\% - 10\%$



Wt. of 135 is 13.50

$135 + 13.5 = 148.5$
 not \$150 that
 started with!!!

